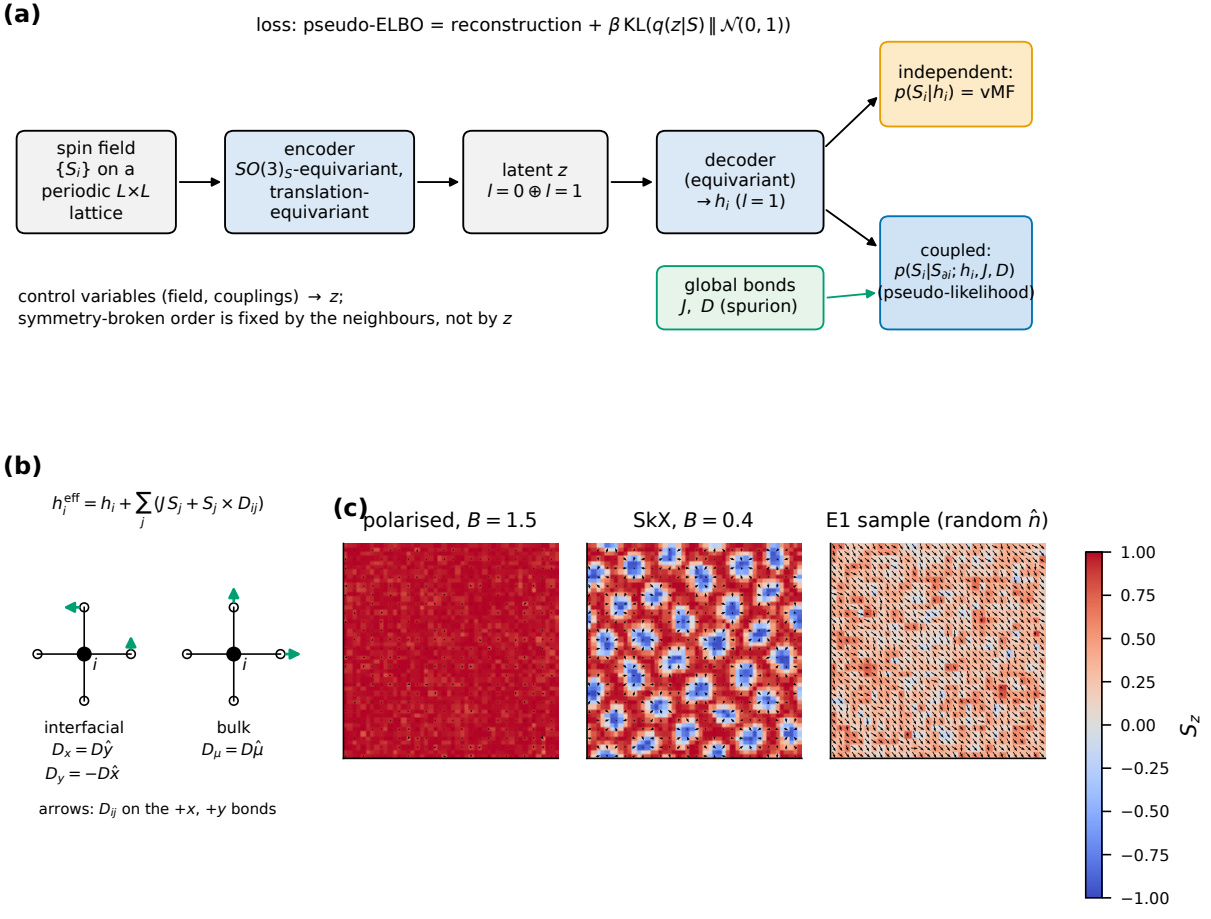

(a)
loss: pseudo-ELBO = reconstruction + $\beta\,\mathrm{KL}(q(z|S)\,\|\,\mathcal{N}(0,1))$
spin field $\{S_i\}$ on a periodic $L\times L$ lattice
encoder $SO(3)_S$-equivariant, translation-equivariant
latent z $l=0\oplus l=1$
decoder (equivariant) $\to h_i$ $(l=1)$
independent: $p(S_i|h_i)$ = vMF
coupled: $p(S_i|S_{\partial i};h_i,J,D)$ (pseudo-likelihood)
global bonds $J,\ D$ (spurion)
control variables (field, couplings) → z;
symmetry-broken order is fixed by the neighbours, not by z
(b)
$h_i^{\mathrm{eff}} = h_i + \sum_j (J\,S_j + S_j \times D_{ij})$
i
i
interfacial
$D_x = D\hat{y}$
$D_y = -D\hat{x}$
bulk
$D_\mu = D\hat{\mu}$
arrows: D_{ij} on the $+x$, $+y$ bonds
(c)
polarised, $B = 1.5$
SkX, $B = 0.4$
E1 sample (random $\hat{n}$)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
S_z